6.Product Sales Analysis

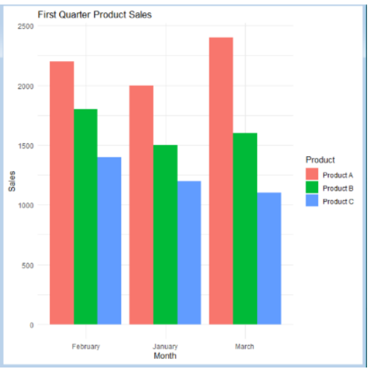
Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

R-code:

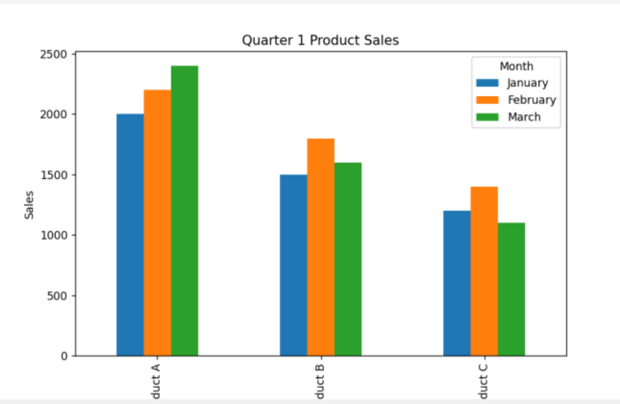
```
ggplot(sales_long, aes(x = Month, y = Sales, fill = Product_Name)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "First Quarter Product Sales",
    x = "Month",
    y = "Sales",
    fill = "Product"
  ) +
  theme_minimal()
```



Python code:

```
sales.set_index("Product").plot(
    kind="bar",
    figsize=(8,5)
)
```

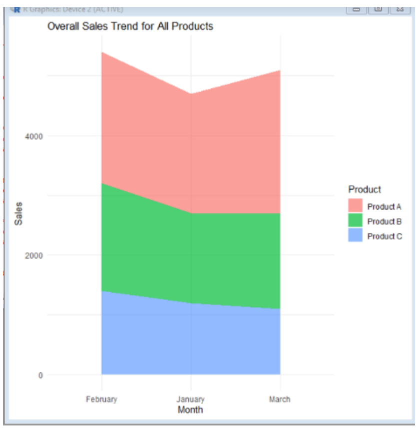
```
plt.title("Quarter 1 Product Sales")
plt.xlabel("Product Name")
plt.ylabel("Sales")
plt.legend(title="Month")
plt.show()
```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

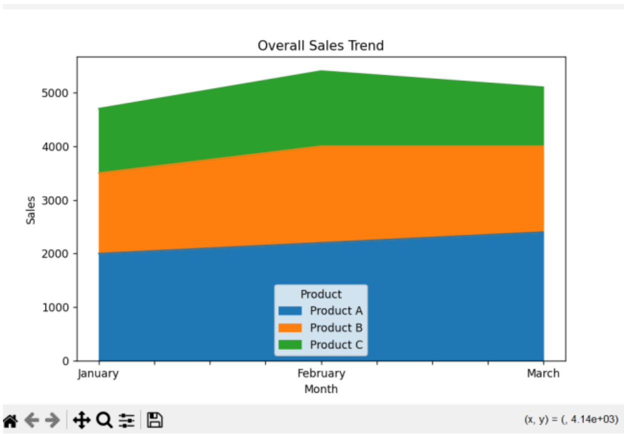
R-code:

```
ggplot(sales_long, aes(x = Month, y = Sales, fill = Product_Name, group = Product_Name)) +
  geom_area(alpha = 0.7) +
  labs(
    title = "Overall Sales Trend for All Products",
    x = "Month",
    y = "Sales",
    fill = "Product"
  ) +
  theme_minimal()
```



Python code:

```
area = sales.set_index("Product").T

area.plot(
    kind="area",
    stacked=True,
    figsize=(8,5)
)

plt.title("Overall Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```



3. Build a table to show the monthly sales data for each product. Label the table elements.

```
sales <- data.frame(
  ProductID = c(1,2,3),
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
```

```
)
sales
print(sales)
```

```
  ProductID   Product January February March
1         1 Product A    2000     2200  2400
2         2 Product B    1500     1800  1600
3         3 Product C    1200     1400  1100
```

Set 7 Customer Demographics Analysis

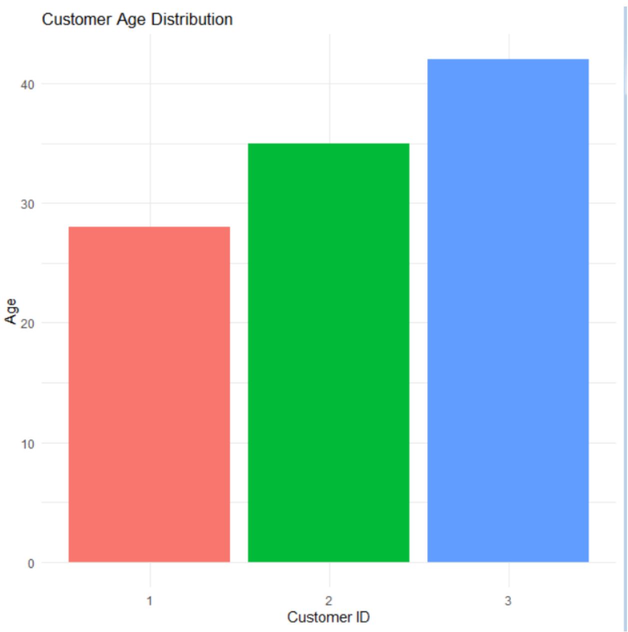
Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in $)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

R -code:

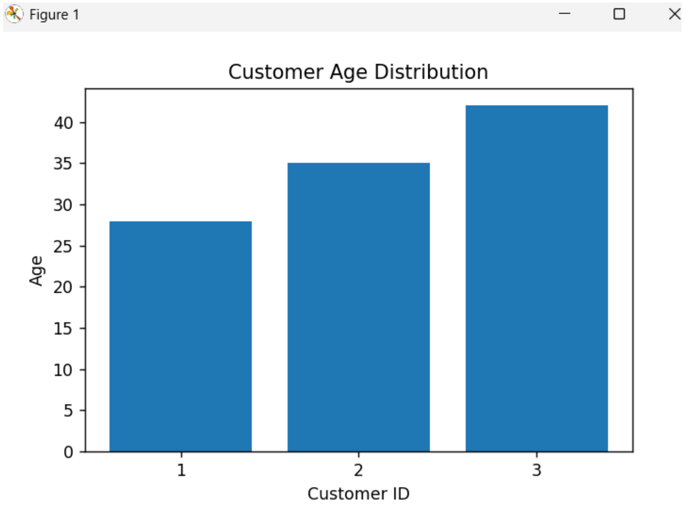
```
ggplot(customer,
    aes(x=factor(CustomerID),
      y=Age,
      fill=factor(CustomerID)))+
  geom_bar(stat="identity")+
  labs(
    title="Customer Age Distribution",
    x="Customer ID",
    y="Age"
  )+
  theme_minimal()+
  theme(legend.position="none")
```



python code-

```
plt.figure(figsize=(6,4))
```

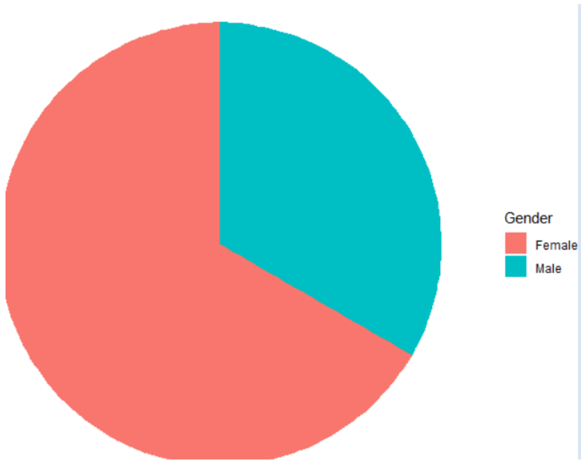
```
plt.bar(customer["Customer ID"].astype(str),
        customer["Age"])

plt.title("Customer Age Distribution")
plt.xlabel("Customer ID")
plt.ylabel("Age")
plt.show()
```



2. Generate a pie chart to represent the distribution of customers by gender.

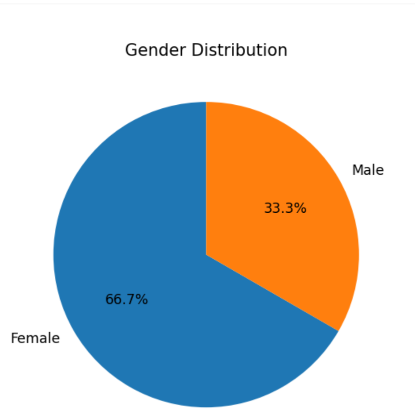
R code-

```
gender_data <- as.data.frame(table(customer$Gender))
colnames(gender_data) <- c("Gender","Count")

ggplot(gender_data,
       aes(x="", y=Count, fill=Gender))+
  geom_bar(stat="identity", width=1)+
  coord_polar("y")+
  labs(title="Gender Distribution")+
  theme_void()
```



python code:

```
gender_count = customer["Gender"].value_counts()

plt.figure(figsize=(5,5))

plt.pie(gender_count,
        labels=gender_count.index,
        autopct="%1.1f%%",
        startangle=90)

plt.title("Gender Distribution")
plt.show()
```



3. Build a table to show the demographic information for each customer. Label the table elements.

```
  CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35   Male  60000
3          3  42 Female  75000
> 
```

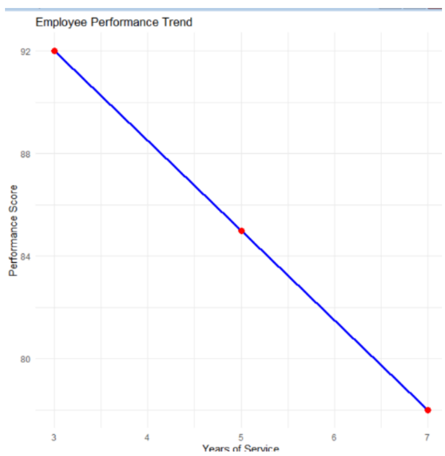
Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

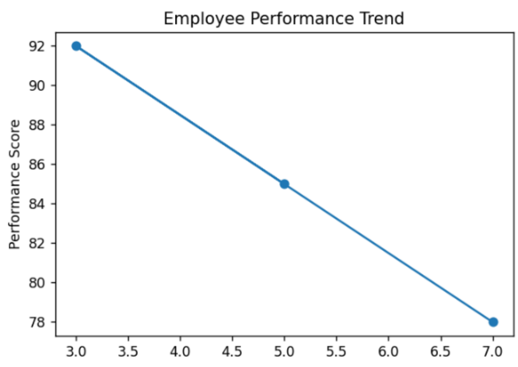
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R code-




```
ggplot(employee,
       aes(x=YearsService,
           y=PerformanceScore,
           group=1))+
  geom_line(color="blue", linewidth=1.2)+
  geom_point(color="red", size=3)+
  labs(
    title="Employee Performance Trend",
    x="Years of Service",
    y="Performance Score"
  )+
  theme_minimal()
```

python code:



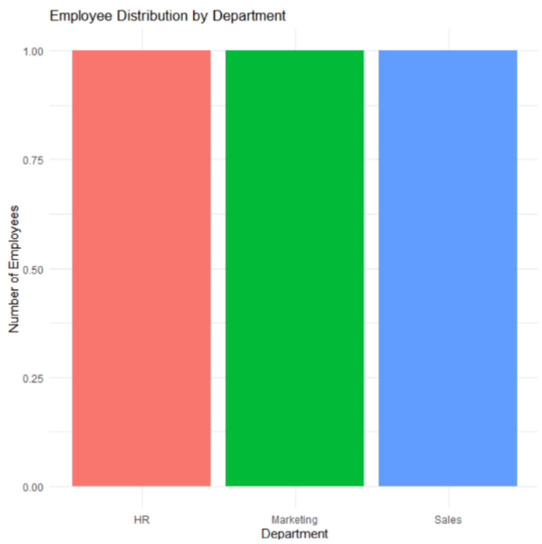

```
plt.figure(figsize=(6,4))

plt.plot(employee["Years of Service"],
        employee["Performance Score"],
        marker="o")
```

```
plt.title("Employee Performance Trend")
plt.xlabel("Years of Service")
plt.ylabel("Performance Score")

plt.show()
```

2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

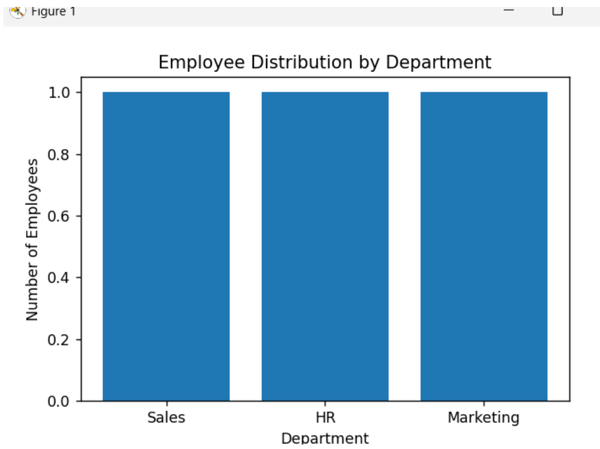
R code-

```
ggplot(employee,
      aes(x=Department,
          fill=Department))+
  geom_bar()+
  labs(
    title="Employee Distribution by Department",
    x="Department",
    y="Number of Employees"
  )+
  theme_minimal()+
  theme(legend.position="none")
```

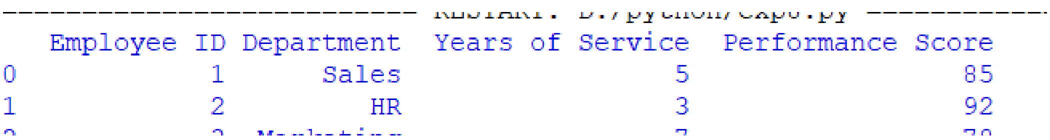


python code:

```
department_count = employee["Department"].value_counts()

plt.figure(figsize=(6,4))

plt.bar(department_count.index,
        department_count.values)

plt.title("Employee Distribution by Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")

plt.show()
```



3. Build a table to display the performance data for each employee. Label the table elements.



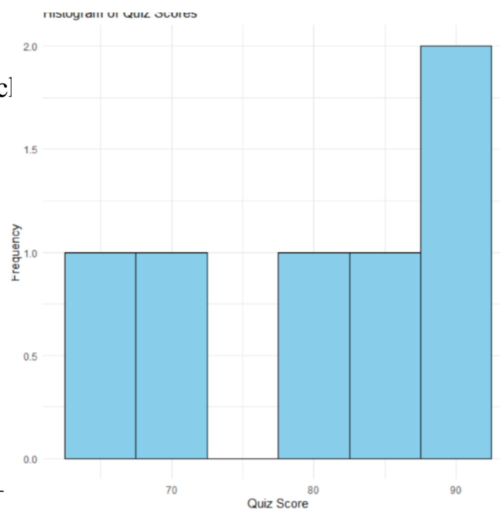
set 9

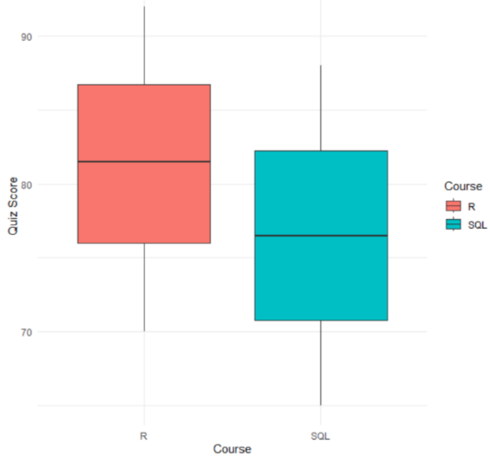
Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

1. Import the dataset in **R**. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

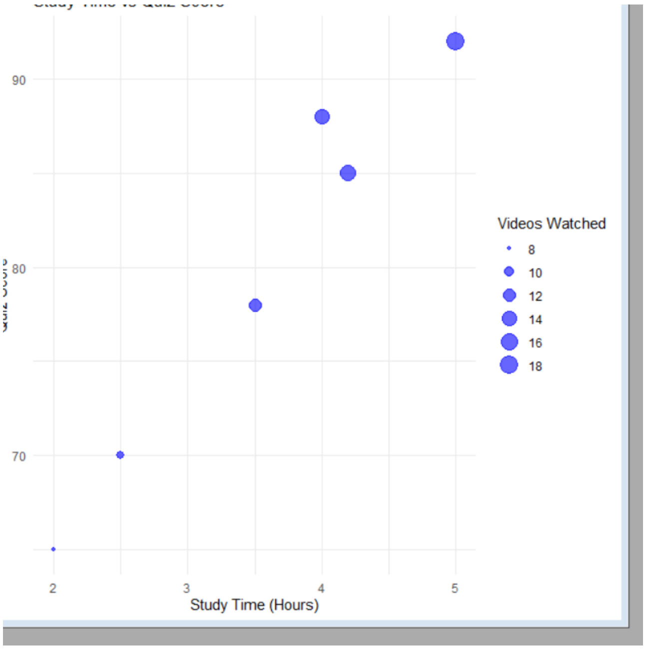
```
ggplot(data, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "blacl
  labs(
    title = "Histogram of Quiz Scores",
    x = "Quiz Score",
    y = "Frequency"
  ) +
  theme_minimal()
ggplot(data, aes(x = Course, y = Quiz_Score, fill = Course)) +
  geom_boxplot() +
  labs(
    title = "Quiz Score by Course",
    x = "Course",
    y = "Quiz Score"
  ) +
  theme_minimal()
```





2. Create a scatter plot **using R programming** between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
ggplot(data,
    aes(x = Study_Time,
      y = Quiz_Score,
      size = Videos_Watched)) +
  geom_point(alpha = 0.6, color = "blue") +
  labs(
    title = "Study Time vs Quiz Score",
    x = "Study Time (Hours)",
    y = "Quiz Score",
    size = "Videos Watched"
  ) +
  theme_minimal()
```



3. Using **R programming**, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```
# Load ggplot2
library(ggplot2)
# Plot Line Chart with Moving Average
ggplot(monthly_avg, aes(x=Month, y=Quiz_Score, group=1)) +
  geom_line(color="blue", linewidth=1.2) +
  geom_point(size=3, color="red") +
  geom_line(aes(y=Moving_Average),           color="darkgreen",
        linewidth=1.2,
        linetype="dashed") +
  labs(
    title="Average Monthly Quiz Score",
    x="Month",
    y="Average Quiz Score"
```

```
print(monthly_avg)
    Month Quiz_Score
2025-01        81.5
2025-02        78.5
2025-03        79.0
```

) +

theme_minimal()

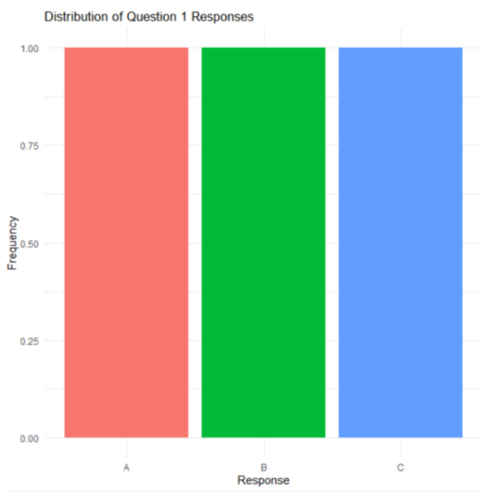
Set 10 Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

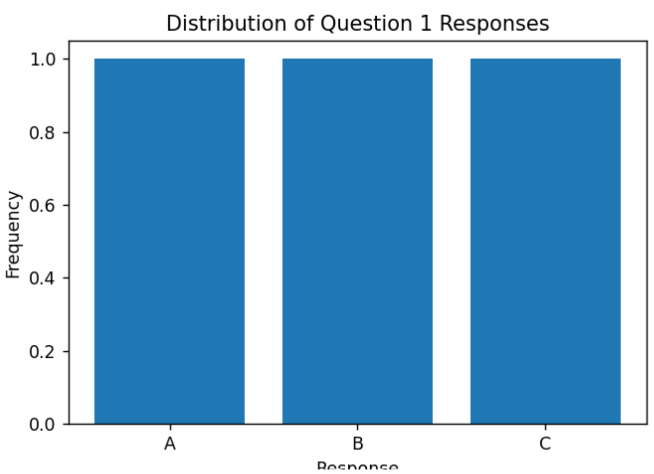
1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

R -code




```
q1_data <- as.data.frame(table(survey$Question1))
colnames(q1_data) <- c("Answer", "Count")

ggplot(q1_data, aes(x = Answer, y = Count, fill = Answer)) +
  geom_bar(stat = "identity") +  labs(
    title = "Distribution of Question 1 Responses",
    x = "Response",
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

python code-



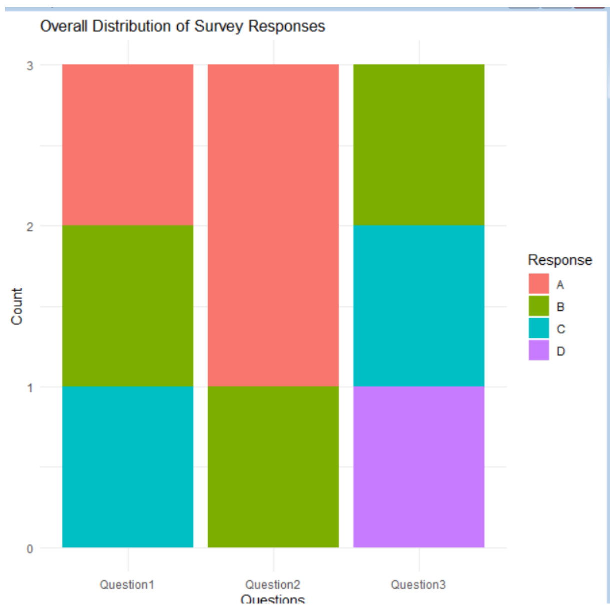

```
q1 = survey["Question 1"].value_counts()

plt.figure(figsize=(6,4))
plt.bar(q1.index, q1.values)
plt.title("Distribution of Question 1 Responses")
plt.xlabel("Response")
plt.ylabel("Frequency")
```

```
plt.show()
```

2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

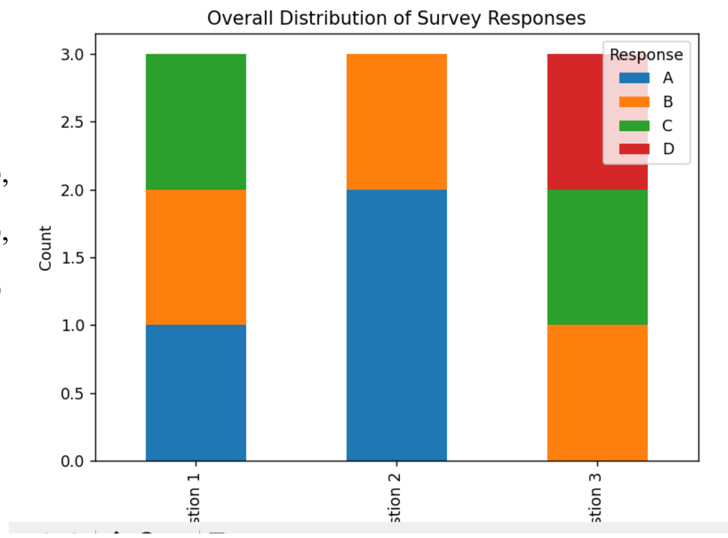
R code-

```
survey_long <- melt(survey,
                    id.vars = "Survey_ID",
                    variable.name = "Question",
                    value.name = "Response")

ggplot(survey_long,
       aes(x = Question,
           fill = Response)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
    y = "Count"
  ) +
  theme_minimal()
```



Python code-

```
response_counts = pd.DataFrame({
    "Question 1": survey["Question 1"].value_counts(),
    "Question 2": survey["Question 2"].value_counts(),
    "Question 3": survey["Question 3"].value_counts()
}).fillna(0)

response_counts.T.plot(
    kind="bar",
    stacked=True,
    figsize=(7,5)
)

plt.title("Overall Distribution of Survey Responses")
```



```
plt.xlabel("Questions")
plt.ylabel("Count")
plt.legend(title="Response")
plt.show()
```

3. Build a table to show the survey response data for each survey. Label the table elements.

	Survey_ID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B